NRF24L01 WIRELESS Communication

SIMULATION · WEB SERIAL API · REAL HARDWARE I/O · 2.4 GHZ

⚙ SIMULATION ONLY | 🔌 HARDWARE ONLY | ↔ BRIDGE MODE

🔌 HARDWARE SERIAL CONNECTIONS – Web Serial API (Chrome/Edge only)

TX ARDUINO – TRANSMITTER PORT

115200 baud | ▶ CONNECT TX

● Not connected

RX ARDUINO – RECEIVER PORT

115200 baud | ▶ CONNECT RX

● Not connected

// NRF24L01 → ARDUINO WIRING (BOTH BOARDS IDENTICAL)

NRF24L01 PIN	ARDUINO PIN	NOTES
VCC	3.3V	⚠ NOT 5V – use 3.3V only (add 10µF cap if unstable)
GND	GND	Common ground
CE	D9	Chip Enable (TX/RX mode select)
CSN	D10	Chip Select Not (SPI)
SCK	D13	SPI Clock
MOSI	D11	SPI Data TX→module
MISO	D12	SPI Data module→TX
IRQ	D2 (opt)	Interrupt – optional

● LINK ACTIVE | FREQ: 2.476 GHz | CHANNEL: 76 | DATA RATE: 1 Mbps

PA LEVEL: HIGH | MODE: HARDWARE

Arduino UNO #1 — HW

📤 TRANSMITTER

NRF24L01 · TX Mode

● PWR ● SPI ● TX ● ACK ● HW

// NRF24L01 PIN MAP

VCC	3.3V ⚠
CE	D9
CSN	D10
SCK	D13
MOSI	D11
MISO	D12

Serial Monitor – TX [HARDWARE CONNECTED]

2.4 GHz RF LINK

RSSI: -57 dBm

Arduino UNO #2 — HW

📥 RECEIVER

NRF24L01 · RX Mode

● PWR ● SPI ● RX ● ACK ● HW

// NRF24L01 PIN MAP

VCC	3.3V ⚠
CE	D9
CSN	D10
SCK	D13
MOSI	D11
MISO	D12

Serial Monitor – RX [HARDWARE CONNECTED]

```
ACK received ✓ (4ms)
[11:57:34.047] >> Packet #5
Payload : "Hello World"
Size : 11 bytes
Channel : 76
CE HIGH → SPI write → CE LOW
ACK received ✓ (6ms)
```

```
Latency : ~4 ms
ACK sent ✓
[11:57:34.861] << Packet #5 received
Data : "Hello World"
RSSI : -57 dBm
Latency : ~6 ms
ACK sent ✓
```

PACKET TRAVEL

MESSAGE PAYLOAD (TX→HW): Hello World

PRESET MESSAGES: -- Select --

CHANNEL (0–125): 76

INTERFERENCE: None

▶ TRANSMIT | ⟳ AUTO | ✕ CLEAR

PACKETS SENT	RECEIVED	DROPPED	SUCCESS RATE	LATENCY
5	5	0	100.0%	6 ms

BYTES SENT	HW PACKETS IN	HW PACKETS OUT
55	0	0

RAW HARDWARE SERIAL I/O – send raw commands to connected Arduinos

→ TX Arduino | Type command (e.g. LED_ON, PING, SERVO:90) | ↑ SEND

📋 ARDUINO SKETCHES – Upload these to your boards

TX Sketch | RX Sketch | Sensor TX